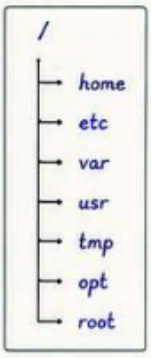
Linux Command Cheat Sheet

1. Linux Directory Structure

/ - Root directory of the system
/home - User home directories
/etc - System configuration files
/var - Variable files (logs, spool, cache)
/usr - User installed programs & data
/tmp - Temporary files
/opt - Optional add-on applications
/root - Home directory for root user



2. Navigation Commands

Command	Description	Example
pwd	Print working directory	pwd
ls	List files and directories	ls
ls -l	Long listing format	ls -l
ls -a	List all files (including hidden)	ls -a
cd <dir>	Change directory	cd /etc
cd ..	Go one directory back	cd ..
cd ~	Go to home directory	cd ~
cd -	Go to previous directory	cd -
clear	Clear terminal screen	clear
history	Show command history	history

3. File & Directory Commands

Command	Description	Example
touch <file>	Create an empty file	touch a.txt
mkdir <dir>	Create a directory	mkdir mydir
mkdir -p <dir>	Create nested directories	mkdir -p a/b/c
rmdir <dir>	Remove empty directory	rmdir emptydir
rm <file>	Remove a file	rm file.txt
rm -r <dir>	Remove directory (recursively)	rm -r mydir
rm -rf <dir>	Force remove directory	rm -rf mydir
cp <src> <dest>	Copy file	cp a.txt b.txt
cp -r <src> <dest>	Copy directory recursively	cp -r mydir bakdir
mv <src> <dest>	Move / Rename file or directory	mv old.txt new.txt
tree	Show directory tree (if installed)	tree

4. File Viewing Commands

Command	Description	Example
cat <file>	Display file content	cat file.txt
less <file>	View file page by page	less file.txt
more <file>	View file page by page (older)	more file.txt
head <file>	Show first 10 lines	head file.txt
head -n <num> <file>	Show first n lines	head -n 20 file.txt
tail <file>	Show last 10 lines	tail file.txt
tail -n <num> <file>	Show last n lines	tail -n 20 file.txt
tail -f <file>	Follow / watch file (live logs)	tail -f app.log
nl <file>	Number all lines	nl file.txt

5. Search Commands

Command	Description	Example
find <path> -name <name>	Search files in a directory	find /home -name test.txt
locate <name>	Search file using database	locate test.txt
which <cmd>	Show path of a command	which python
whereis <cmd>	Show binary, source, man info	whereis java
grep <pattern> <file>	Search pattern in a file	grep "error" app.log
grep -r <pattern> <dir>	Search pattern in directory	grep -r "fail" /var/log

grep Example :

grep "error" app.log → Search the word "error" in app.log

6. Permissions

chmod <permissions> <file/dir>

r (read) = 4
w (write) = 2
x (execute) = 1

Examples :

chmod 755 script.sh → rwxr-xr-x
chmod 644 file.txt → rw-r--r--
chmod 777 file.txt → rwxrwxrwx

Permission Table

Type	r	w	x	Value
Read	Yes	-	-	4
Write	-	Yes	-	2
Execute	-	-	Yes	1

Permission Representation

Owner	Group	Others
rwx	r-x	r-x
7	5	5

7. Ownership

Command	Description	Example
chown <user> <file/dir>	Change owner	chown aman file.txt
chgrp <group> <file/dir>	Change group	chgrp developers file.txt
chown <user>:<group> <file/dir>	Change owner and group	chown aman:dev file.txt

8. File Compression

Command	Description	Example
zip <file.zip> <files>	Create zip file	zip files.zip file.txt
unzip <file.zip>	Extract zip file	unzip files.zip
tar -cvf <file.tar> <files>	Create tar file	tar -cvf data.tar data/
tar -xvf <file.tar>	Extract tar file	tar -xvf data.tar
tar -czvf <file.tar.gz> <dir>	Create tar.gz (compressed)	tar -czvf logs.tar.gz logs/
tar -xzvf <file.tar.gz>	Extract tar.gz file	tar -xzvf logs.tar.gz